

## Day 13 MYSQL ASSIGNMENT

```
USE slatask;

CREATE TABLE dept13(
dept_id INT,
dept_name VARCHAR(50)
);

CREATE TABLE emp13(
emp_id INT,
emp_name VARCHAR(50),
dept_id INT,
manager_id INT,
salary INT
);

CREATE TABLE proj13(
project_id INT,
project_name VARCHAR(50),
dept_id INT
);

CREATE TABLE emp_proj13(
emp_id INT,
project_id INT
);

CREATE TABLE loc13(
location_id INT,
dept_id INT,
city VARCHAR(50)
);

INSERT INTO dept13 VALUES
(1,'IT'),
(2,'HR'),
(3,'Finance'),
(4,'Sales');

INSERT INTO emp13 VALUES
(1,'Ravi',1,NULL,82000),
(2,'Kumar',1,1,62000),
(3,'Priya',2,1,52000),
```

```

(4,'Arjun',3,2,72000),
(5,'Meena',NULL,2,48000);
INSERT INTO proj13 VALUES
(201,'Inventory',1),
(202,'Accounts',3),
(203,'Hiring',2);
INSERT INTO emp_proj13 VALUES

```

```

(1,201),
(2,201),
(3,203),
(4,202);

```

```

INSERT INTO loc13 VALUES
(1,1,'Chennai'),
(2,2,'Delhi'),
(3,3,'Mumbai');

```

-- 1 Employees earning more than average salary

```

SELECT *
FROM emp13
WHERE salary > (SELECT AVG(salary) FROM emp13);

```

| Result Grid                       |        |          |         |            |        |
|-----------------------------------|--------|----------|---------|------------|--------|
| Filter Rows: <input type="text"/> |        |          |         |            |        |
| Export: <input type="text"/>      |        |          |         |            |        |
|                                   | emp_id | emp_name | dept_id | manager_id | salary |
| ▶                                 | 1      | Ravi     | 1       | NULL       | 82000  |
|                                   | 4      | Arjun    | 3       | 2          | 72000  |

-- 2 Employees in IT department

```

SELECT *
FROM emp13
WHERE dept_id =
(SELECT dept_id FROM dept13 WHERE dept_name='IT');

```

| Result Grid                       |        |          |         |            |        |
|-----------------------------------|--------|----------|---------|------------|--------|
| Filter Rows: <input type="text"/> |        |          |         |            |        |
| Export: <input type="text"/>      |        |          |         |            |        |
|                                   | emp_id | emp_name | dept_id | manager_id | salary |
|                                   | 1      | Ravi     | 1       | NULL       | 82000  |
|                                   | 2      | Kumar    | 1       | 1          | 62000  |

-- 3 Employees with highest salary

```

SELECT *
FROM emp13
WHERE salary =
(SELECT MAX(salary) FROM emp13);

```

|   | emp_id | emp_name | dept_id | manager_id | salary |
|---|--------|----------|---------|------------|--------|
| ▶ | 1      | Ravi     | 1       | NULL       | 82000  |

-- 4 Employees without department

```

SELECT *
FROM emp13
WHERE dept_id IS NULL;

```

|   | emp_id | emp_name | dept_id | manager_id | salary |
|---|--------|----------|---------|------------|--------|
| ▶ | 5      | Meena    | NULL    | 2          | 48000  |

-- 5 Employees working on at least one project

```

SELECT *
FROM emp13
WHERE emp_id IN
(SELECT emp_id FROM emp_proj13);

```

|   | emp_id | emp_name | dept_id | manager_id | salary |
|---|--------|----------|---------|------------|--------|
| ▶ | 1      | Ravi     | 1       | NULL       | 82000  |
|   | 2      | Kumar    | 1       | 1          | 62000  |
|   | 3      | Priya    | 2       | 1          | 52000  |
|   | 4      | Arjun    | 3       | 2          | 72000  |

-- 6 Employees NOT working on project

```

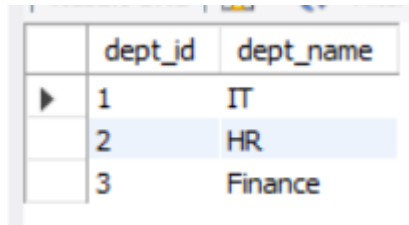
SELECT *
FROM emp13
WHERE emp_id NOT IN
(SELECT emp_id FROM emp_proj13);

```

|   | emp_id | emp_name | dept_id | manager_id | salary |
|---|--------|----------|---------|------------|--------|
| ▶ | 5      | Meena    | NULL    | 2          | 48000  |

-- 7 Departments having employees

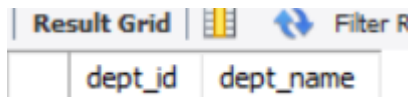
```
SELECT *  
FROM dept13  
WHERE dept_id IN  
(SELECT dept_id FROM emp13);
```



|   | dept_id | dept_name |
|---|---------|-----------|
| ▶ | 1       | IT        |
|   | 2       | HR        |
|   | 3       | Finance   |

-- 8 Departments without employees

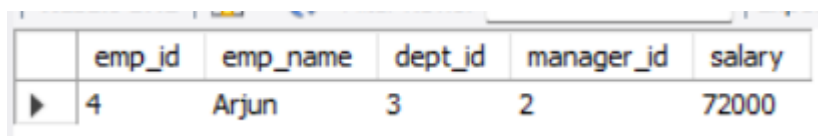
```
SELECT *  
FROM dept13  
WHERE dept_id NOT IN  
(SELECT dept_id FROM emp13);
```



|  | dept_id | dept_name |
|--|---------|-----------|
|--|---------|-----------|

-- 9 Employees earning more than manager

```
SELECT *  
FROM emp13 e1  
WHERE salary >  
(SELECT salary FROM emp13 e2  
WHERE e1.manager_id=e2.emp_id);
```



|   | emp_id | emp_name | dept_id | manager_id | salary |
|---|--------|----------|---------|------------|--------|
| ▶ | 4      | Arjun    | 3       | 2          | 72000  |

-- 10 Employees in Mumbai departments

```
SELECT *  
FROM emp13  
WHERE dept_id IN  
(SELECT dept_id FROM loc13 WHERE city='Mumbai');
```

|   | emp_id | emp_name | dept_id | manager_id | salary |
|---|--------|----------|---------|------------|--------|
| ▶ | 4      | Arjun    | 3       | 2          | 72000  |

-- 11 Employees working on Inventory project

```
SELECT *
FROM emp13
WHERE emp_id IN
(SELECT emp_id FROM emp_proj13
WHERE project_id=
(SELECT project_id FROM proj13
WHERE project_name='Inventory'));
```

|   | emp_id | emp_name | dept_id | manager_id | salary |
|---|--------|----------|---------|------------|--------|
| ▶ | 1      | Ravi     | 1       | NULL       | 82000  |
|   | 2      | Kumar    | 1       | 1          | 62000  |

-- 12 Projects assigned to IT employees

```
SELECT *
FROM proj13
WHERE dept_id=
(SELECT dept_id FROM dept13
WHERE dept_name='IT');
```

|   | project_id | project_name | dept_id |
|---|------------|--------------|---------|
| ▶ | 201        | Inventory    | 1       |

-- 13 Employees who are managers

```
SELECT *
FROM emp13
WHERE emp_id IN
(SELECT manager_id FROM emp13
WHERE manager_id IS NOT NULL);
```

|   | emp_id | emp_name | dept_id | manager_id | salary |
|---|--------|----------|---------|------------|--------|
| ▶ | 1      | Ravi     | 1       | NULL       | 82000  |
|   | 2      | Kumar    | 1       | 1          | 62000  |

-- 14 Employees NOT managers

```
SELECT *  
FROM emp13  
WHERE emp_id NOT IN  
(SELECT manager_id FROM emp13  
WHERE manager_id IS NOT NULL);
```

|   | emp_id | emp_name | dept_id | manager_id | salary |
|---|--------|----------|---------|------------|--------|
| ▶ | 3      | Priya    | 2       | 1          | 52000  |
|   | 4      | Arjun    | 3       | 2          | 72000  |
|   | 5      | Meena    | NULL    | 2          | 48000  |

-- 15 Departments having projects

```
SELECT *  
FROM dept13  
WHERE dept_id IN  
(SELECT dept_id FROM proj13);
```

|   | dept_id | dept_name |
|---|---------|-----------|
| ▶ | 1       | IT        |
|   | 2       | HR        |
|   | 3       | Finance   |

-- 16 Departments without projects

```
SELECT *  
FROM dept13  
WHERE dept_id NOT IN  
(SELECT dept_id FROM proj13);
```

|   | dept_id | dept_name |
|---|---------|-----------|
| ▶ | 4       | Sales     |

-- 17 Employees earning more than dept average

```
SELECT *  
FROM emp13 e1  
WHERE salary >  
(SELECT AVG(salary) FROM emp13 e2
```

WHERE e1.dept\_id=e2.dept\_id);

|   | emp_id | emp_name | dept_id | manager_id | salary |
|---|--------|----------|---------|------------|--------|
| ▶ | 1      | Ravi     | 1       | NULL       | 82000  |

-- 18 Second highest salary

SELECT MAX(salary)

FROM emp13

WHERE salary <

(SELECT MAX(salary) FROM emp13);

|   | MAX(salary) |
|---|-------------|
| ▶ | 72000       |

-- 19 Employees in departments with >1 employee

SELECT \*

FROM emp13

WHERE dept\_id IN

(SELECT dept\_id FROM emp13

GROUP BY dept\_id

HAVING COUNT(\*)>1);

|   | emp_id | emp_name | dept_id | manager_id | salary |
|---|--------|----------|---------|------------|--------|
| ▶ | 1      | Ravi     | 1       | NULL       | 82000  |
|   | 2      | Kumar    | 1       | 1          | 62000  |

-- 20 Employees whose department has location

SELECT \*

FROM emp13

WHERE dept\_id IN

(SELECT dept\_id FROM loc13);

|   | emp_id | emp_name | dept_id | manager_id | salary |
|---|--------|----------|---------|------------|--------|
| ▶ | 1      | Ravi     | 1       | NULL       | 82000  |
|   | 2      | Kumar    | 1       | 1          | 62000  |
|   | 3      | Priya    | 2       | 1          | 52000  |
|   | 4      | Arjun    | 3       | 2          | 72000  |

-- 21 Employees whose department has NO location

```
SELECT *  
FROM emp13  
WHERE dept_id NOT IN  
(SELECT dept_id FROM loc13);
```

|  | emp_id | emp_name | dept_id | manager_id | salary |
|--|--------|----------|---------|------------|--------|
|--|--------|----------|---------|------------|--------|

-- 22 Projects belonging to departments with employees

```
SELECT *  
FROM proj13  
WHERE dept_id IN  
(SELECT dept_id FROM emp13);
```

|   | project_id | project_name | dept_id |
|---|------------|--------------|---------|
| ▶ | 201        | Inventory    | 1       |
|   | 202        | Accounts     | 3       |
|   | 203        | Hiring       | 2       |

-- 23 Employees working in own department project

```
SELECT *  
FROM emp13 e  
WHERE emp_id IN  
(SELECT emp_id FROM emp_proj13 ep  
JOIN proj13 p  
ON ep.project_id=p.project_id  
WHERE e.dept_id=p.dept_id);
```

|   | emp_id | emp_name | dept_id | manager_id | salary |
|---|--------|----------|---------|------------|--------|
| ▶ | 1      | Ravi     | 1       | NULL       | 82000  |
|   | 2      | Kumar    | 1       | 1          | 62000  |
|   | 3      | Priya    | 2       | 1          | 52000  |
|   | 4      | Arjun    | 3       | 2          | 72000  |

-- 24 Employees working outside department project

```
SELECT *
```



```

FROM emp13 e
WHERE emp_id IN
(SELECT emp_id FROM emp_proj13 ep
JOIN proj13 p
ON ep.project_id=p.project_id
WHERE e.dept_id<>p.dept_id);

```

|             |                                                                                   |                                                                                   |              |            |        |
|-------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|--------------|------------|--------|
| Result Grid |  |  | Filter Rows: |            | Expo   |
|             | emp_id                                                                            | emp_name                                                                          | dept_id      | manager_id | salary |

-- 25 Departments all employees salary >50000

```

SELECT *
FROM dept13
WHERE dept_id IN
(SELECT dept_id FROM emp13
GROUP BY dept_id
HAVING MIN(salary)>50000);

```

|   | dept_id | dept_name |
|---|---------|-----------|
| ▶ | 1       | IT        |
|   | 2       | HR        |
|   | 3       | Finance   |